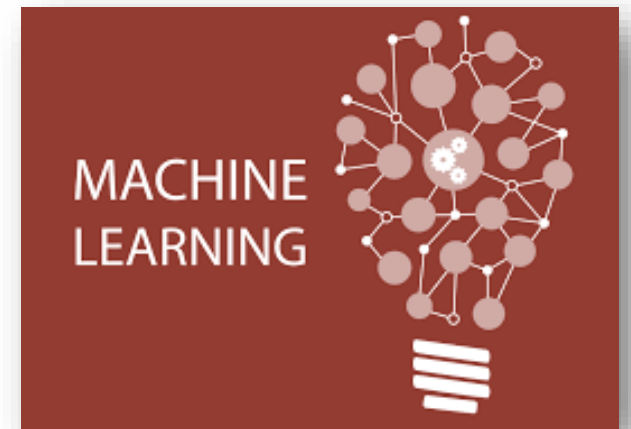




Machine Learning

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The COOLEST TOPIC IN SCIENCE

- **“A breakthrough in machine learning would be worth ten Microsoft” (Bill Gates, chairman, Microsoft)**
- **“Machine learning is the next internet” (tony tether, director, DARPA)**
- **machine learning is the hot new thing” (John Hennessey, president, Stanford)**
- **“Web rankings today are mostly a matter of machine learning” (Prabhakar Raghavan, dir. Research, Yahoo)**
- **“machine learning is going to result in a real revolution” (Greg Papadopoulos, CTO, Sun)**



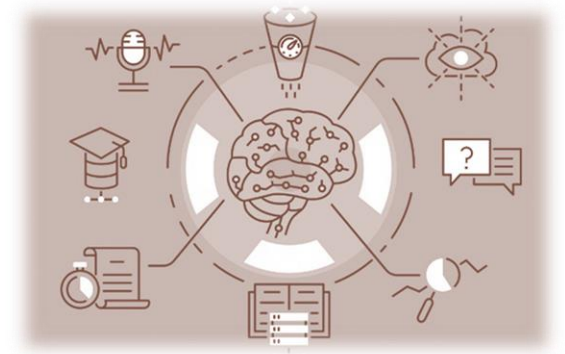
What is Machine Learning?

- machine learning is the science (and art) of programming computers so they can *learn from data*.
- The field of study that gives computers the ability to learn without being explicitly programmed.

—*Arthur Samuel, 1959*

- A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

—*Tom Mitchell, 1997*





Problems Machine Learning Can Solve

- Human expertise does not exist (navigating on mars)



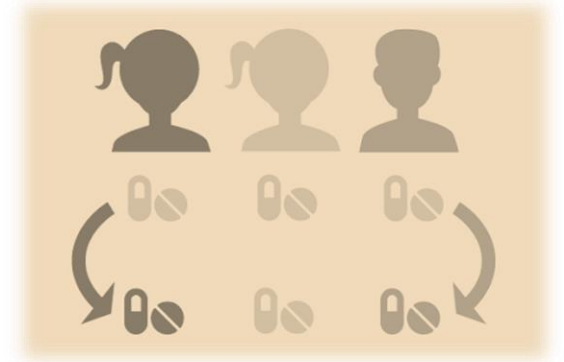
- Humans can't explain their expertise (speech recognition)





Problems Machine Learning Can Solve

- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



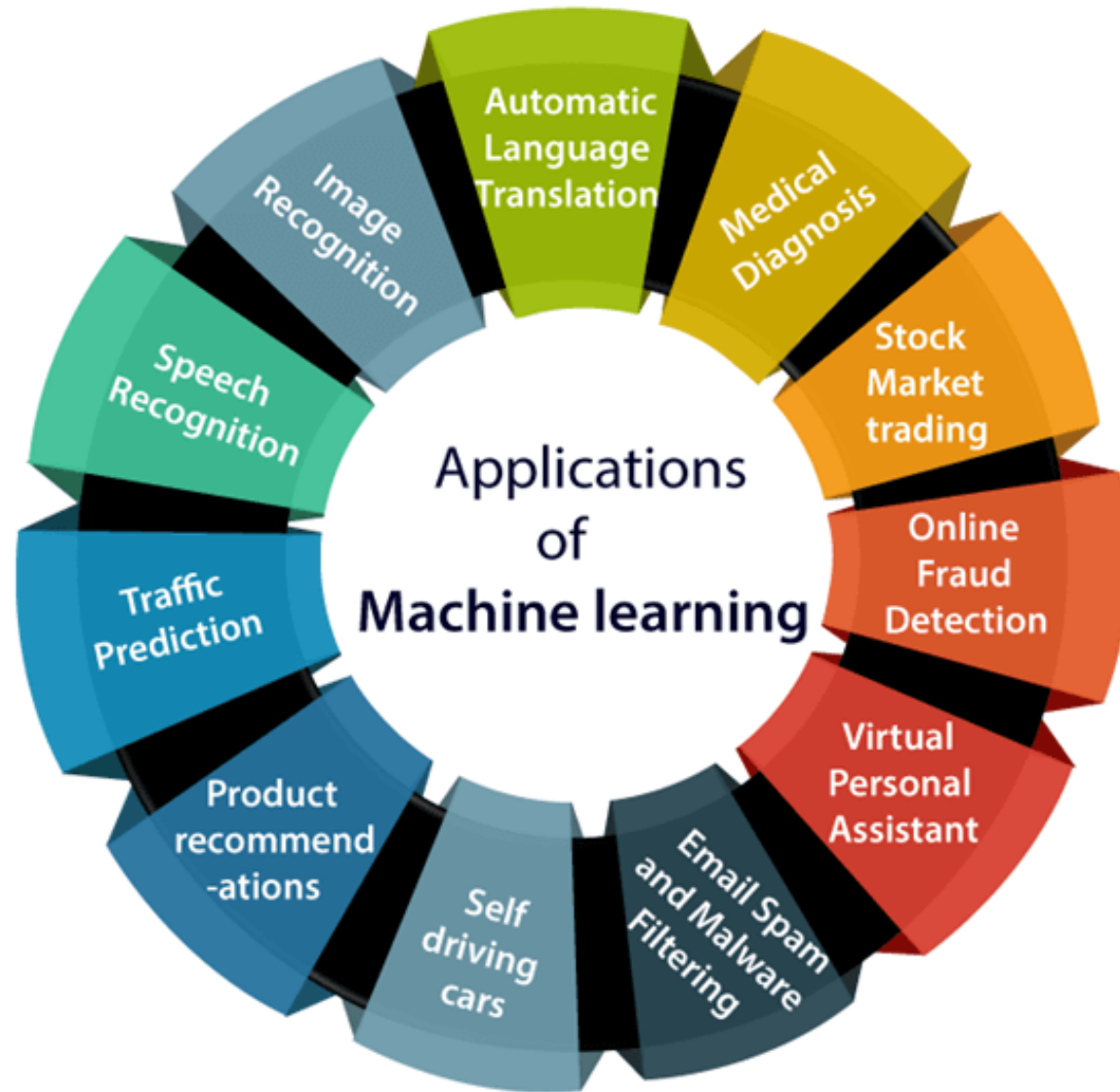




Image Recognition:

Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **automatic friend tagging suggestion**:





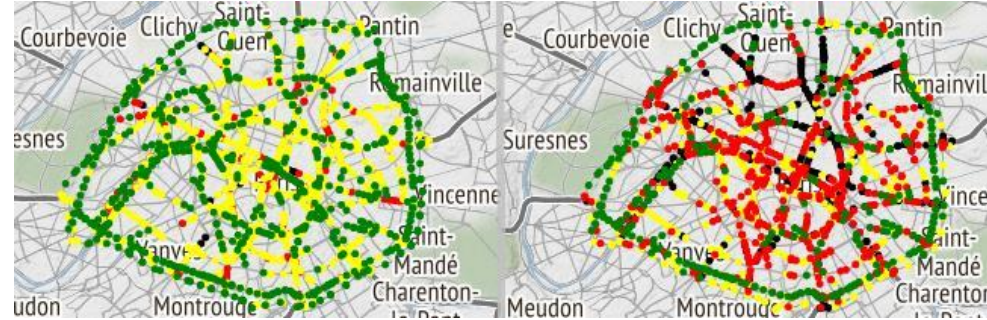
Speech Recognition

While using google, we get an option of "search by voice," it comes under speech recognition, and it's a popular application of machine learning.

Speech recognition is a process of converting voice instructions into text, and it is also known as "speech to text", or "computer speech recognition." At present, machine learning algorithms are widely used by various applications of speech recognition. Google assistant, Siri, Cortana, and Alexa are using speech recognition technology to follow the voice instructions.



Traffic prediction:



If we want to visit a new place, we take help of google maps, which shows us the correct path with the shortest route and predicts the traffic conditions.

It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

- **Real time location** of the vehicle from google map app and sensors
- **Average time has taken** on past days at the same time.



Product recommendations:

amazon

NETFLIX

Machine learning is widely used by various e-commerce and entertainment companies such as **amazon**, **Netflix**, etc., For product recommendation to the user. Whenever we search for some product on amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

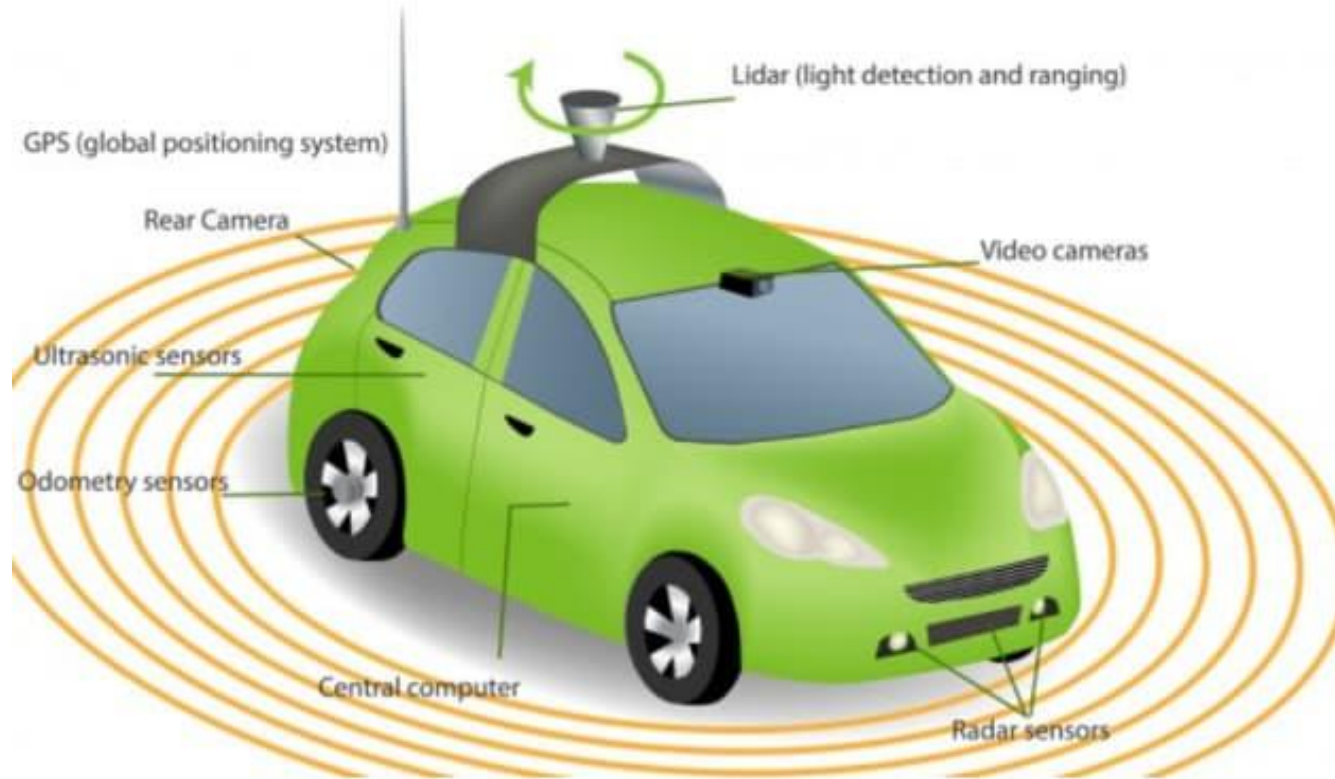
Google understands the user interest using various machine learning algorithms and suggests the product as per customer interest.



Self-driving cars (Autonomous Cars):



Self-driving cars (Autonomous Cars):





Email Spam and Malware Filtering:





Online Fraud Detection:





Stock Market trading:





Medical Diagnosis:





Automatic Language Translation:

